203736: porphyritic metagranitic rock, Victory deposit

(Black Flag Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)

Location and sampling

Widgiemooltha (SH 51-14), Lake Lefroy (3235)
MGA Zone 51, 383970E 6532837N

Sampled on 2 September 2020

This sample was collected from the 488.75 – 493.00 m depth interval of drillcore CD20297-W1, drilled in 2019 by St Ives Gold Mining Company Pty Limited at their Victory deposit (Nixon, 2019), with support from the Westerm Australian Government's Exploration Incentive Scheme (EIS). The drillhole is located on the eastern shore of Lake Lefroy, about 19.2 km east of Pilbailey Hill, 17.6 south-southeast of Kambalda, and 10.0 km west-southwest of Parker Hill.

Geological context

The unit sampled is a porphyritic metagranitic rock (Fig. 1), known informally as the 'Flames porphyry', and assigned to the 2692–2655 Ma Black Flag Group of the Kalgoorlie Terrane (Smithies et al., 2018; Hall et al., 2019). The Black Flag Group consists mainly of felsic metavolcaniclastic and metasedimentary rocks with subordinate felsic and mafic volcanic rocks. The Black Flag Group overlies the 2753–2680 Ma Kalgoorlie Group, is unconformably overlain by the 2664–2640 Ma Merougil Formation (Hall et al., 2019), and in part is intruded by the c. 2680 Ma Golden Mile Dolerite (Rasmussen et al., 2009). The sample was collected to verify its assignment to the Black Flag Group. A porphyritic meta-igneous rock, sampled from drillhole NFD003 about 3.1 km to the northeast, yielded an igneous crystallization age of 2681 ± 5 Ma (GSWA 112110, Nelson, 1995). A porphyritic hornblende metagranodiorite, sampled from drillhole LD7006 about 4.6 km to the north-northwest, yielded an igneous crystallization age of 2666 ± 7 Ma (GSWA 229913, Lu et al., 2020a). A porphyritic microgranite, sampled from drillhole LD81635 about 13.0 km to the north-northwest, yielded an igneous crystallization age of 2682 ± 14 Ma (GSWA 230328, Lu et al., 2020b).

Petrographic description

The sample is a fine- to medium-grained, sparsely porphyritic metagranitic rock (Fig. 1), consisting of plagioclase, quartz, biotite, and minor carbonate minerals, chlorite, titanite, epidote, magnetite, apatite, sericite and anhydrite. Plagioclase and quartz phenocrysts are distributed within a fine-grained, recrystallized, quartzofeldspathic groundmass. Plagioclase phenocrysts are subhedral to euhedral, partially zoned, and up to 1 cm in size. Quartz phenocrysts are anhedral, embayed, and up to 2 mm in size. Anhedral to subhedral pseudomorphs of former hornblende phenocrysts are up to 1 cm in size and composed of fine-grained biotite laths and minor carbonate minerals. Coarser flakes of biotite in the groundmass are partially altered to chlorite. Titanite, epidote and secondary magnetite are closely associated with biotite.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and subhedral to euhedral. The crystals are up to 200 µm long, and elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and some zircons appear to contain older cores. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed over three sessions, on 9–10, 15–16, and 22–23 October 2020, using SHRIMP-B. Analyses 1.1 to 8.1 (spot numbers 1–8) were obtained during the first session, together with 13 analyses of the M257 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.24% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the first session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.39%. Analyses 9.1 to 5.2 (spot numbers 9–16) were obtained during the second session, together with 16 analyses of the M257 standard, of which 14 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.28% (1σ). Analyses 15.1 to 26.1 (spot numbers 17–28) were obtained during the third session, together with 12 analyses of the M257 standard, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.31% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Twenty-eight analyses were obtained from 26 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

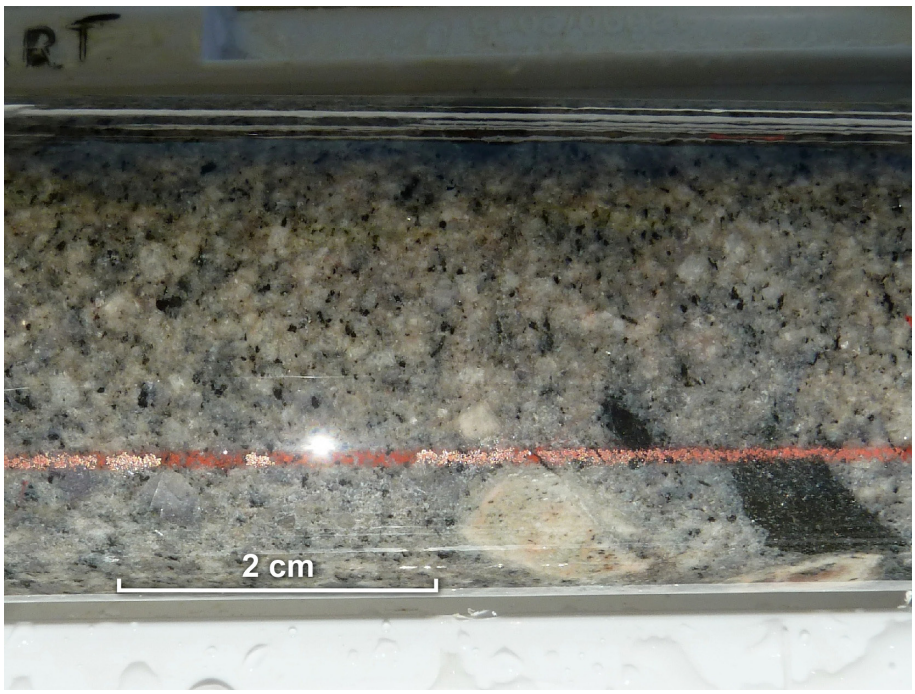

Figure 1. **Drillcore image (wet surface) of sample 203736: porphyritic metagranitic rock, Victory deposit**

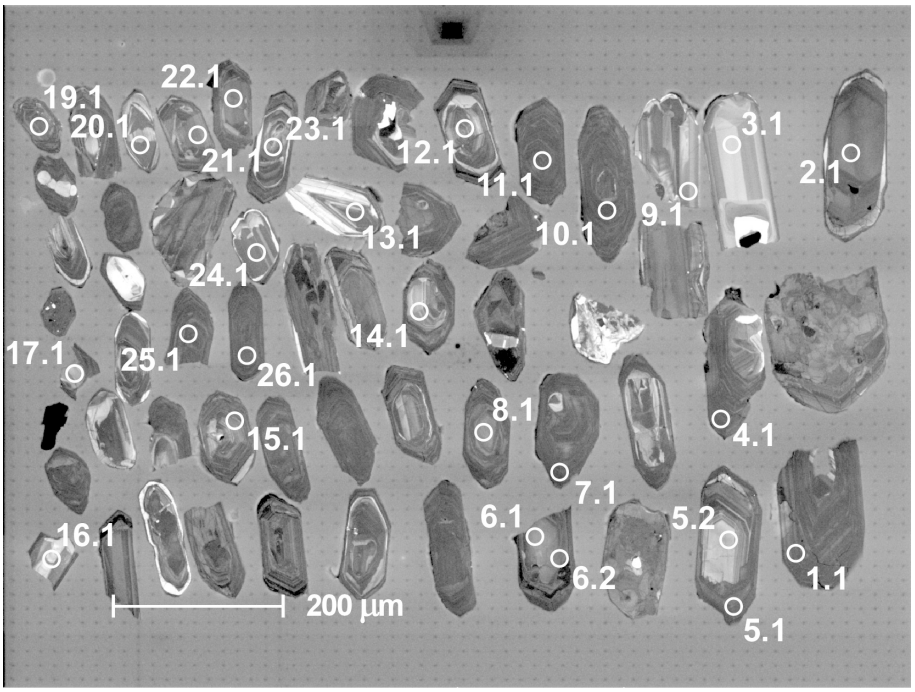

Figure 2. **Cathodoluminescence image of all zircons from sample 203736: porphyritic metagranitic rock, Victory deposit. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 203736: porphyritic metagranitic rock, Victory deposit

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	20	18.1	14	0	0.01	0.849	1.877	0.085	0.18559	0.00553	1.893	0.086	0.17801	0.00777	2734	102	2634	73	-3.8
I	15	6.2	106	44	0.41	0.000	2.017	0.036	0.18011	0.00193	2.017	0.036	0.18011	0.00193	2596	38	2654	18	2.2
I	22	20.1	104	92	0.89	0.044	1.854	0.031	0.18066	0.00175	1.855	0.031	0.18028	0.00179	2780	38	2655	16	-4.7
I	16	5.2	42	17	0.41	-0.136	1.945	0.055	0.17955	0.00324	1.942	0.055	0.18076	0.00345	2677	62	2660	32	-0.7
I	5	5.1	263	54	0.21	-0.065	2.015	0.023	0.18061	0.00132	2.013	0.023	0.18120	0.00135	2600	24	2664	12	2.4
I	11	11.1	329	76	0.23	-0.016	2.024	0.024	0.18111	0.00111	2.023	0.024	0.18126	0.00112	2589	25	2664	10	2.8
I	12	12.1	183	96	0.53	0.030	1.995	0.029	0.18296	0.00149	1.995	0.029	0.18270	0.00152	2619	31	2678	14	2.2
I	6	6.1	62	27	0.44	0.216	1.911	0.038	0.18479	0.00240	1.915	0.038	0.18286	0.00265	2708	44	2679	24	-1.1
I	17	15.1	78	29	0.38	-0.119	1.926	0.059	0.18236	0.00208	1.924	0.059	0.18342	0.00221	2698	68	2684	20	-0.5
I	26	24.1	120	91	0.76	-0.130	2.013	0.033	0.18241	0.00177	2.010	0.033	0.18357	0.00189	2603	35	2685	17	3.1
I	23	21.1	139	34	0.24	-0.075	2.039	0.031	0.18304	0.00165	2.038	0.031	0.18371	0.00171	2574	33	2687	15	4.2
I	2	2.1	126	30	0.24	0.059	2.039	0.029	0.18428	0.00176	2.040	0.029	0.18375	0.00180	2572	30	2687	16	4.3
I	3	3.1	24	24	1.01	-0.545	1.873	0.057	0.17963	0.00367	1.863	0.057	0.18451	0.00460	2770	69	2694	41	-2.8
I	13	13.1	208	179	0.86	0.026	1.976	0.027	0.18505	0.00139	1.977	0.027	0.18482	0.00141	2639	30	2697	13	2.1
I	14	14.1	57	25	0.43	0.000	2.027	0.046	0.18508	0.00259	2.027	0.046	0.18508	0.00259	2585	48	2699	23	4.2
I	24	22.1	126	32	0.26	-0.209	1.945	0.031	0.18411	0.00176	1.941	0.031	0.18597	0.00194	2678	36	2707	17	1.1
I	18	16.1	82	40	0.49	-0.111	1.975	0.036	0.18508	0.00203	1.973	0.036	0.18607	0.00215	2644	40	2708	19	2.4
I	9	9.1	64	53	0.83	-0.170	1.985	0.043	0.18486	0.00249	1.982	0.043	0.18638	0.00271	2634	47	2711	24	2.8
X	8	8.1	90	44	0.49	-0.464	1.867	0.032	0.19309	0.00206	1.858	0.032	0.19725	0.00243	2776	39	2804	20	1.0
P	1	1.1	274	82	0.30	-0.030	2.107	0.023	0.17783	0.00129	2.106	0.023	0.17810	0.00130	2504	23	2635	12	5.0
P	4	4.1	372	93	0.25	0.058	2.031	0.021	0.17899	0.00119	2.032	0.021	0.17847	0.00122	2580	22	2639	11	2.2
P	28	26.1	1359	925	0.68	0.033	2.039	0.018	0.18072	0.00053	2.039	0.018	0.18043	0.00054	2572	19	2657	5	3.2
D	27	25.1	945	430	0.46	0.060	2.248	0.030	0.16915	0.00067	2.249	0.030	0.16861	0.00069	2371	27	2544	7	6.8
D	7	7.1	368	73	0.20	0.131	2.272	0.023	0.17708	0.00122	2.275	0.023	0.17592	0.00128	2348	20	2615	12	10.2
D	10	10.1	600	267	0.44	0.074	2.118	0.021	0.17892	0.00078	2.120	0.021	0.17826	0.00081	2491	20	2637	8	5.5
D	21	19.1	190	90	0.47	-0.146	2.073	0.030	0.18244	0.00148	2.070	0.030	0.18374	0.00159	2540	30	2687	14	5.5
D	25	23.1	125	42	0.34	-0.043	2.147	0.097	0.18392	0.00177	2.146	0.097	0.18430	0.00181	2466	93	2692	16	8.4
D	19	17.1	33	4	0.13	-0.144	2.067	0.057	0.18787	0.00329	2.064	0.057	0.18916	0.00353	2547	58	2735	31	6.9

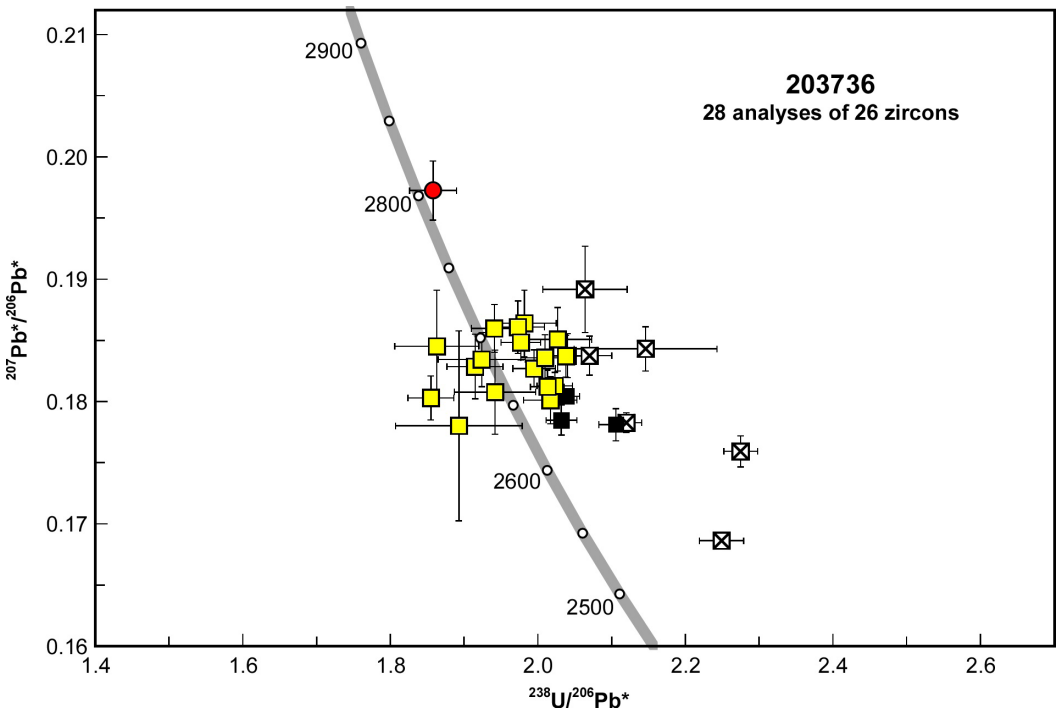

Figure 3. U–Pb analytical data for sample 203736: porphyritic metagranitic rock, Victory deposit. Yellow squares indicate Group I (magmatic zircons); red circle indicates Group X (xenocrystic zircon core); black squares indicate Group P (radiogenic-Pb loss); crossed squares indicate Group D (discordance >5%)

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Six analyses are >5% discordant. The dates obtained from these six analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 22 analyses can be divided into three groups, based on their ^{207}Pb*/^{206}Pb* ratios.

Group I comprises 18 analyses of 16 zircons (Table 1), which yield a weighted mean ^{207}Pb*/^{206}Pb* date of 2680 ± 8 Ma (MSWD = 1.1).

Group X comprises one analysis of a zircon core (Table 1), which yields a ^{207}Pb*/^{206}Pb* date of 2804 ± 20 Ma (1σ).

Group P comprises three analyses of three zircons (Table 1), which yield ^{207}Pb*/^{206}Pb* dates of 2657–2635 Ma.

The date of 2680 ± 8 Ma for the 18 analyses in Group I is interpreted as the age of igneous crystallization. The date of 2804 ± 20 Ma (1σ) for the single analysis in Group X is interpreted as the age of an inherited zircon. The dates of 2657–2635 Ma for the three analyses in Group P are interpreted to reflect loss of radiogenic Pb

References

Hall, CE, Wyche, NL and Sapkota, J (partial report), Black Flag Group (A-BF-xf-s): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 05 January 2022, <www.dmp.wa.gov.au/ens>.

Lu, Y, Wingate, MTD and Smithies, RH 2020a, 229913: hornblende metagranodiorite, Lake Lefroy; Geochronology Record 1653: Geological Survey of Western Australia, 4p.

Lu, Y, Wingate, MTD and Smithies, RH 2020b, 230328: porphyritic microgranite, Lake Lefroy; Geochronology Record 1654: Geological Survey of Western Australia, 4p.

Nelson, DR 1995, 112110: felsic volcanic rock, Nelson's Fleet; Geochronology Record 526: Geological Survey of Western Australia, 4p.

Nixon, D 2019, Victory, Final Report, Co-funded Drilling Program DAG2018-01286324, St Ives Gold Mining Company Pty Limited: Exploration Report A121223 (unpublished), 15p.

Rasmussen, B, Mueller, AG and Fletcher, IR 2009, Zirconolite and xenotime U-Pb age constraints on the emplacement of the Golden Mile Dolerite sill and gold mineralization at the Mt Charlotte mine, Eastern Goldfields Province, Yilgarn Craton, Western Australia: Contributions to Mineralogy and Petrology, v. 157, p. 559–572.

Smithies, RH, Lu, Y, Kirkland, CL, Cassidy, KF, Champion, DC, Sapkota, J, De Paoli, M and Burley, L 2018, A new look at lamprophyres and sanukitoids, and their relationship to the Black Flag Group and gold prospectivity: Geological Survey of Western Australia, Record 2018/15, 23p.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH, Cassidy, KF and Smithies, RH 2022, 203736: porphyritic metagranitic rock, Victory deposit; Geochronology Record 1852: Geological Survey of Western Australia, 4p.

Data obtained: 23 October 2020

Data released: 29 April 2022